

CS & IT ENGINEERING

Programming in C

Arrays and Pointers

Lec- 06



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TOPICS TO BE
COVERED



Arrays and Pointers

Gate-2010

Q

A) -9

B) 5

☒ C) 15

D) 19

E) None

```
int f(int *a, int n){
```

```
    if(n <= 0)
```

```
        return 0;
```

```
    else if(*a / 2 == 0)
```

```
        return *a + f(a+1, n-1);
```

```
    else
```

```
        return *a - f(a+1, n-1);
```

```
void main(){
```

```
    int a[] = {12, 7, 13, 4, 11, 6};
```

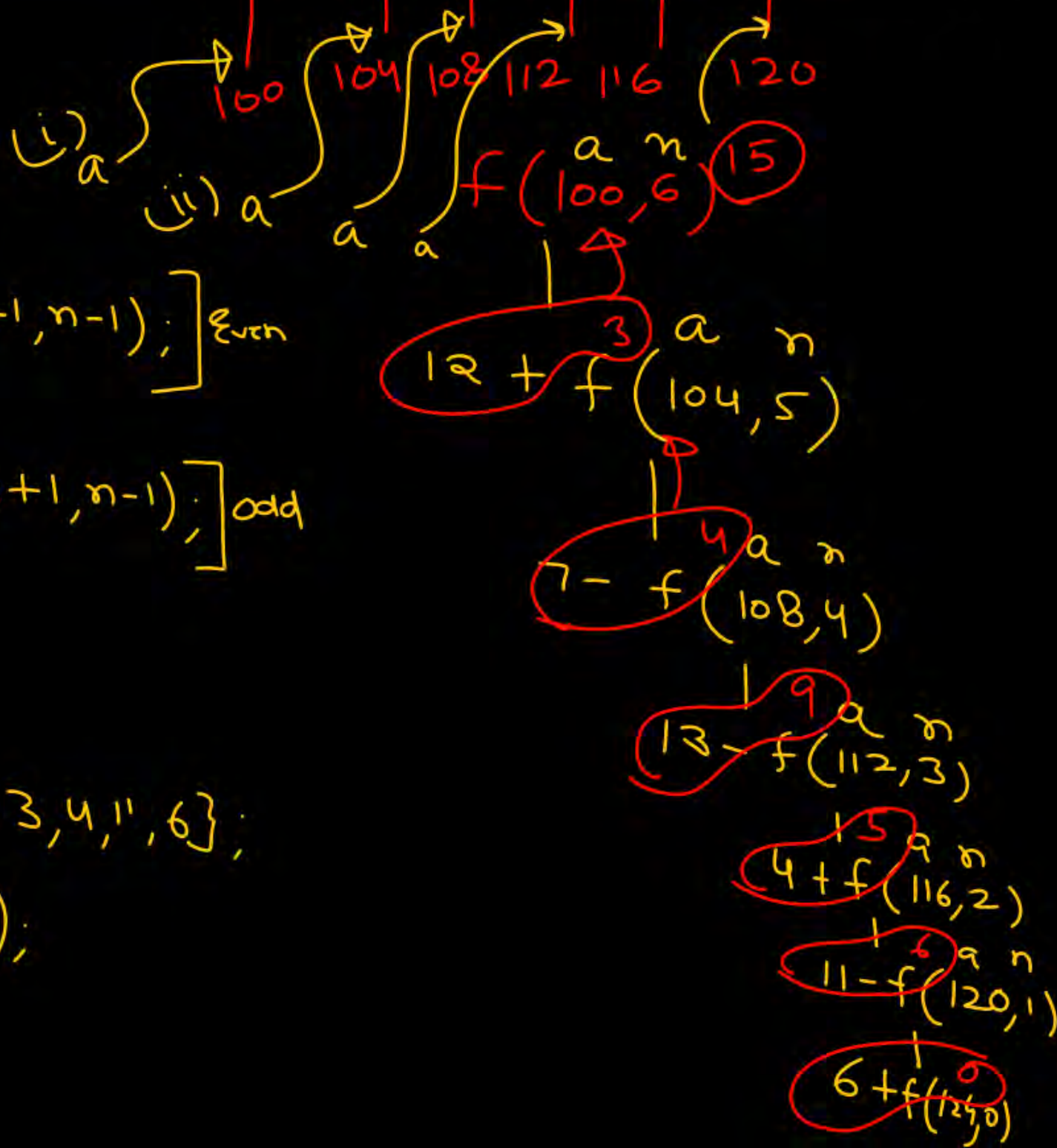
```
    printf("%d", f(a, 6));
```

```
}
```

12 / 2 == 0

a[0] a[1] a[2] a[3] a[4] a[5]

12	7	13	4	11	6
----	---	----	---	----	---



Gate 2016
Q

```
int f(int* p, int n){
```

```
    if(n <= 1) return 0;
```

```
    else
```

```
        return max(f(p+1, n-1), p[0]-p[1]);
    }
```

```
int main(){
```

```
    int a[] = {3, 5, 2, 6, 4};
```

```
    printf("%d", f(a, 5));
```

```
}
```

$2a[0]$ $f(100, 5)$

(i) $p+0 = p$

$*p+0$

$p[0] = 3$
 $p[1] = 5$

p

$a[0]$ $a[1]$ $a[2]$ $a[3]$ $a[4]$

3	5	2	6	4
---	---	---	---	---

100 104 108 112 116

p

p

p

$f(100, 5)$

3

$p[0] = 3, p[1] = 5$

$\max(f(104, 4), 3-5)$

$p[0] = 5$
 $p[1] = 2$

$\max(f(108, 3), 3)$

$p[0] = 2$
 $p[1] = 6$

$\max(f(112, 2), -4)$

$p[0] = 6, p[1] = 4$

$\max(f(116, 1), 2)$

Gate 2019

```
int main() {
```

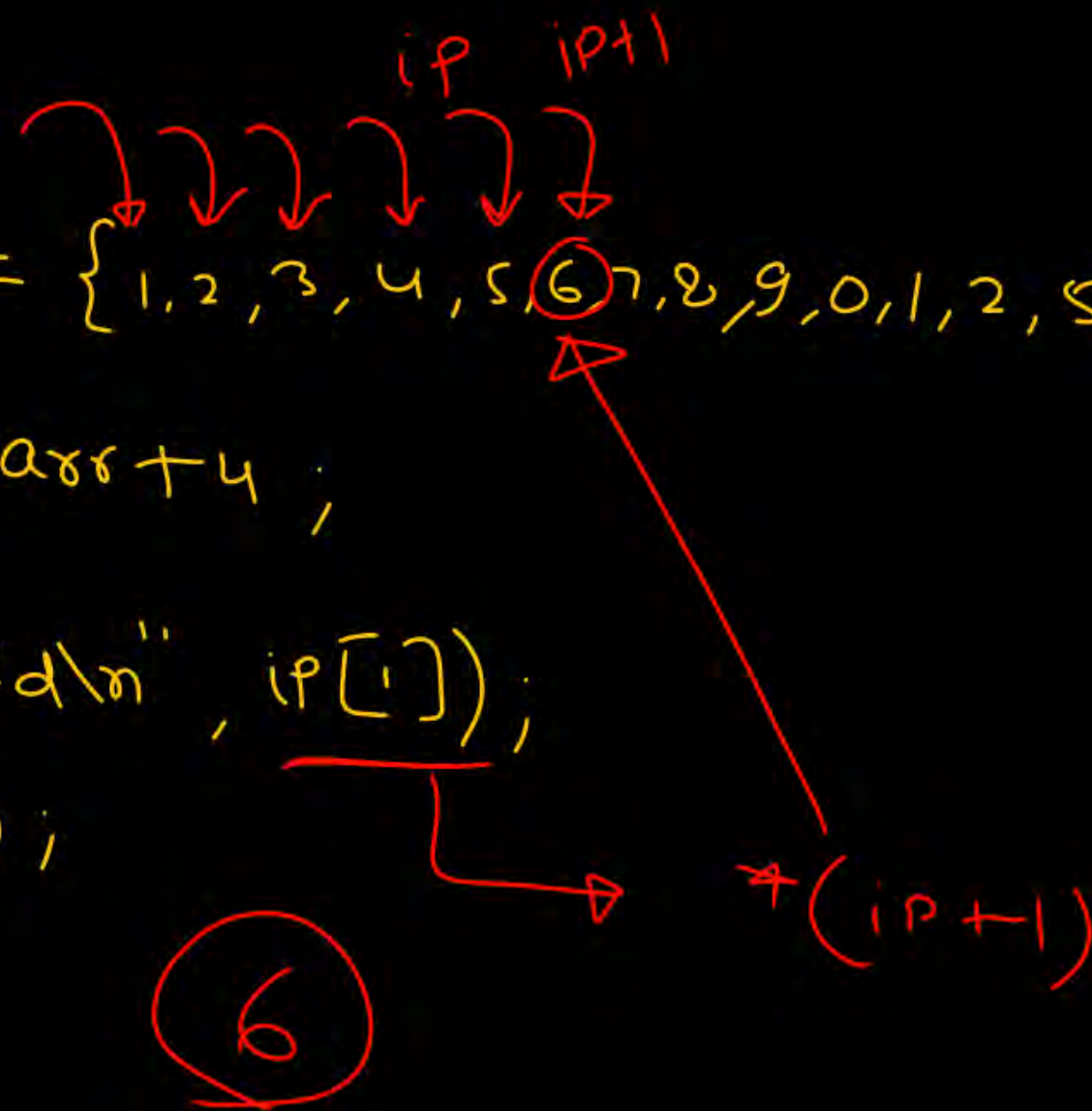
```
int arr[] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 0, 1, 2, 5};
```

```
int *ip = arr + 4;
```

```
printf("%d\n", ip[1]);
```

```
return 0;
```

```
}
```



Gate 2019

```
int a[] = {2, 4, 6, 8, 10};
```

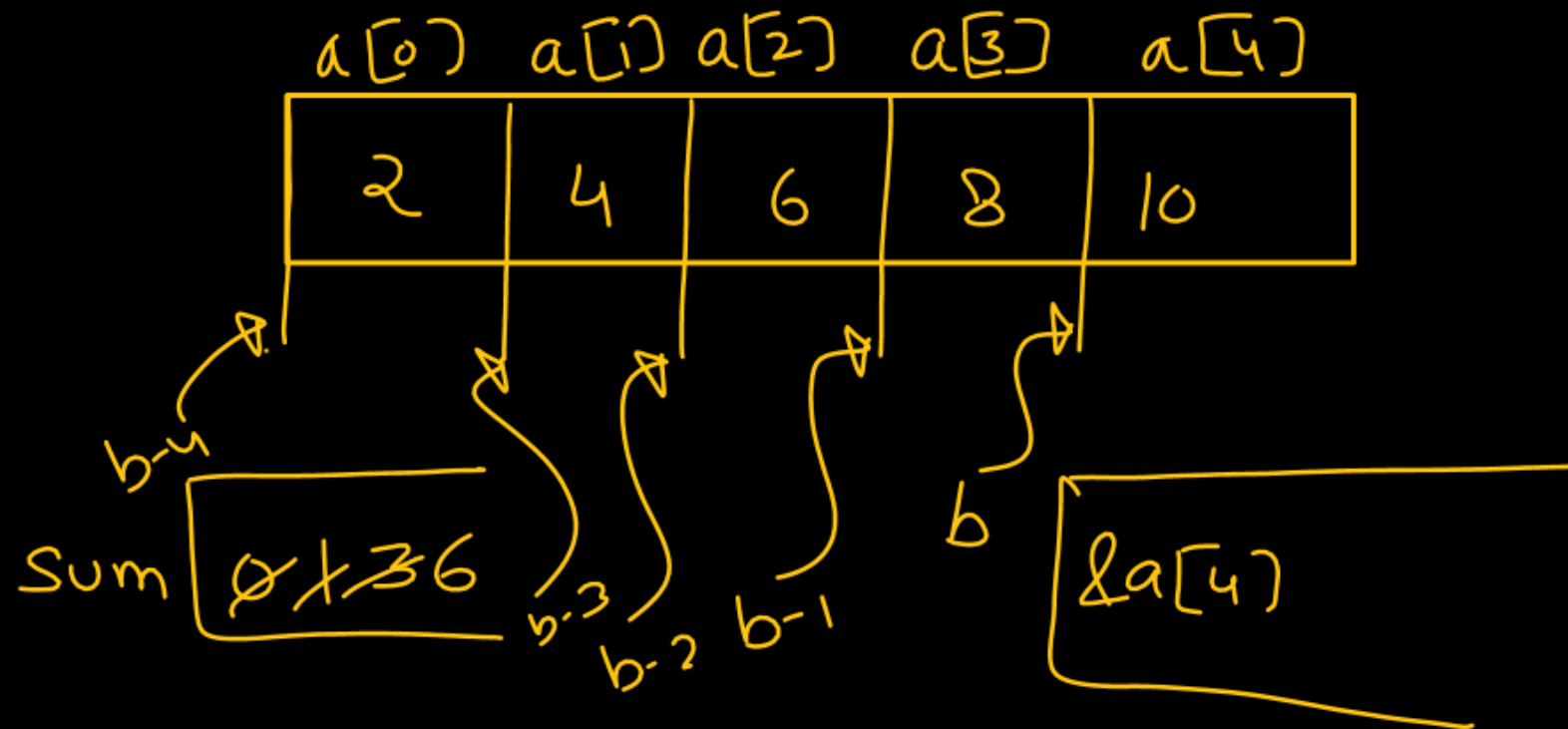
```
int i, sum = 0, *b = a + 4;
```

```
for (i = 0; i < 5; i++)
```

```
    sum = sum + (*b - i) - *b;
```

```
printf("%d\n", sum);
```

10



(i) $i = 0$ $sum = sum + (10 - 0) - 10 = 0$

(ii) $i = 1$ $sum = sum + (10 - 1) - (8)$
 $= 0 + 9 - 8 = 1$

(iii) $i = 2$

$sum = sum + (10 - 2) - 6 = 1 + 8 - 6$
 $= 3$

(iv) $i = 3$ $sum = sum + (10 - 3) - 4 = 3 + 7 - 4 = 6$

(v) $i = 4$, $sum = sum + (10 - 4) - 2 = 6 + 6 - 2 = 10$

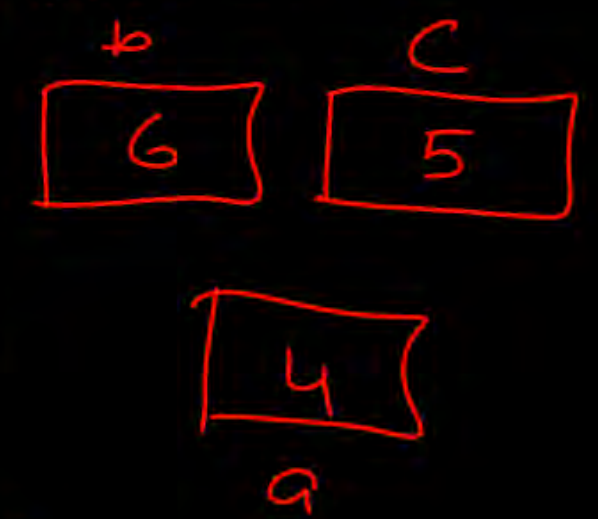
GATE 2015

```
void f1(int a, int b)
{
    int c;
    c = a; a = b; b = c;
}
```

```
void f2(int *a, int *b)
{
    int c;
    c = *a;
    *a = *b;
    *b = c;
}
```

```
int main() {
    int a = 4, b = 5, c = 6;
    f1(a, b);
    f2(&b, &c);
    printf("%d", c - a - b);
}
```

call by value



$$5 - 4 - 6$$

$$= -5$$

Gate 2010

```
void f(int *p, int *q)
```

```
{
```

```
    p = q;
```

```
    *p = 2;
```

```
}
```

```
int i = 0, j = 1;
```

```
int main() {
```

```
    ⇒ f(&i, &j);
```

```
    printf("%d %d\n", i, j);
```

```
    return 0;
```

```
}
```

A) 2 2

B) 2 1

C) 0 1

D) 0 2

MCS

i
0

j
~~1~~ 2

1016

2096

P ~~1016~~ 2096 q 2096

Gate - 2016

Q

```
void f(int *p, int m)
```

```
{
```

```
    m = m + 5;
```

```
    *p = *p + m;  
    return;
```

```
}
```

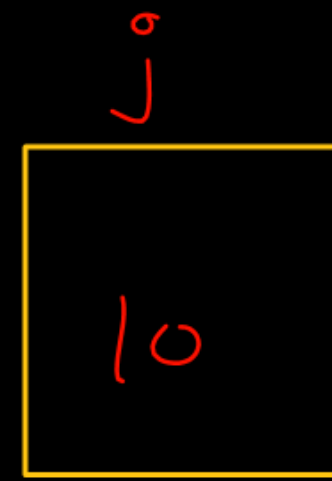
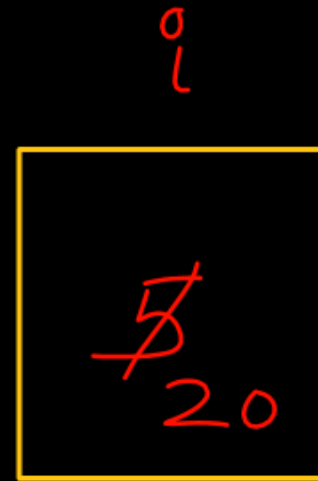
```
void main(){
```

```
    int i = 5, j = 10;
```

```
    f(&i, j);
```

```
    printf("%d", 20+10 i+j);
```

```
}
```



1016

2036

p



m

30

(gate 2016) void mystery(int *Ptr a, int *Ptr b)

{

int *temp;

temp = Ptr b;

Ptr b = Ptr a;

Ptr a = temp;

}

No dereferencing

void main() {

int a = 2016, b = 0, c = 4, d = 42;

mystery(&a, &b);

if(a < c)

mystery(&c, &a);

mystery(&a, &d);

printf("%d", a);

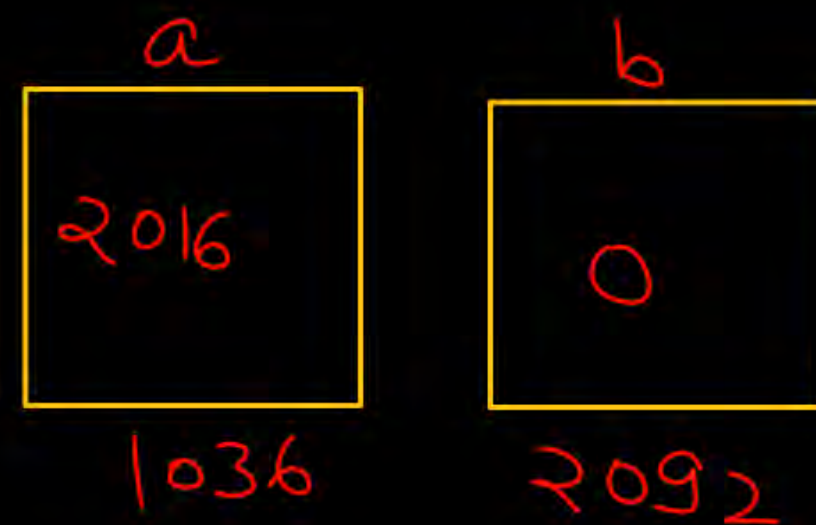
}

(gate 2016)

```
void mystery(int *Ptra, int *Ptrb)
```

```
{  
    int *temp;  
    temp = Ptrb;  
    Ptrb = Ptra;  
    Ptra = temp;  
}
```

```
void main() {  
    int a = 2016, b = 0, c = 4, d = 42;  
    mystery(&a, &b);  
    if (a < c)  
        mystery(&c, &a);  
    mystery(&a, &d);  
    printf("%d", a);  
}
```



Gate - 2020

Q

```
int tob(int b, int* arr)
{
    int i;
    for(i=0; b>0; i++)
    {
        if(b/2) arr[i]=1; odd
        else arr[i]=0;
        b = b/2; Even
    }
    return i;
}
```

(i) $i = 0, 4 > 0$ (iii) $arr[2] = 1$
 $arr[0] = 0$
(ii) $b > 0$ $2 > 0$ (iv) $0 > 0$
 $arr[1] = 0$

arr	0	0	1					-	-
A	0	1	2	3	4	5	6	int	pp (: 1 : 1 : 1 : 1)

908 $\int^v$

b

0	x
4	7

i

3

2

~~0~~

$$\begin{array}{|c|} \hline a \\ \hline 3 \\ \hline \end{array}$$

5
4

i_{pp}

ex

8	1	x	8	1
3	9			

```
int PP(int a, int b)
```

```
{
    int arr[20];
    int i, tot = 1, ex, len;
    ex = a;
```

```
len = tob(b, arr);
for (i = 0; i < len; i++)
```

tot

~~1~~81

```

    }
    return (tot);
}

```

value returned by $PP(3,4)$

(i) $i = 0$
 $arr[0] == 1$
 false

```

(ii) i = 1
    i < 3
    arr[i] ==
    false
    ++ )

```

(iii) $l = 2$
 $2 < 3$
 $arr[2] = 4$
 $\boxed{ex;}$ $\checkmark$ $True$

$$e_x = e_x \times e_x; \quad \checkmark$$

(iv) $i = 3$
353

81

Gate 2006

```
void swap(int *x, int *y)
{
    static int *temp;
    temp = x;
    x = y; y = temp;
}
```

```
void printab() {
    static int i, a = -3, b = -6;
    i = 0;
    while (i <= 4) {
        if ((i++) % 2 == 1) continue;
        a = a + i;
        b = b + i;
    }
    swap(&a, &b);
    printf("a = %d, b = %d\n", a, b);
}
```

A) a = 0, b = 3
a = 0, b = 3

B) a = 3, b = 0
a = 12, b = 9

C) a = 3, b = 6
a = 3, b = 6

☒ D) a = 6, b = 3
a = 15, b = 12

```
void main() {
    printab();
    printab();
}
```


Gate-2009

```
int fun(int n, int *f-p)
{
    int t, f;
    if (n <= 1) {
        *f-p = 1;
        return 1;
    }
```

```
    t = fun(n-1, f-p);
    f = t + *f-p;
    *f-p = t;
    return f;
}
```

```
int main() {
    int x = 15;
```

```
    printf("/d\n", fun(5, &x));
    return 0;
}
```

A) 6

B) 8

C) 14

D) 15

Grade - 2009

```
int fun(int n, int* f-p)
```

```
{
```

```
int t, f;
```

```
if (n <= 1) {
```

```
    *f-p = 1;  
    return 1;  
}
```

```
1. t = fun(n-1, f-p);
```

```
2. f = t + *f-p;
```

```
3. *f-p = t;
```

```
4. return f;
```

```
}
```

5

t = fun(4, 1016)

f = 5 + 3

*f-p = 5

return 8

3

t = fun(3, 1016)

f = 3 + 2 = 5

*f-p = 3

return 5

2

t = fun(2, 1016)

f = 3

*f-p = 2

return 3

1

t = fun(1, 1016)

f = 2

*f-p = 1

return 2

8
fun(5, 1016)

x [15 + 235]
1016

Gate 2006

Useless Printab()

```
void swap(int *x, int *y)
{
    static int *temp;
    temp = x;
    x = y; y = temp;
}
```

```
void main() {
    printab();
    printab();
}
```

i $\boxed{0+2=2}$ $a = -2+3$
 $b = -5+3$
a $\boxed{-3-2=-5}$ (iv) $3 \leq 4$
b $\boxed{-6-5=-11}$ $3/2 = 1$ true
Continue ✓

```
void printab() {
    static int i, a = -3, b = -6;
    i = 0;
    while (i <= 4) {
        if ((i++) / 2 == 1) continue;
        a = a + i;
        b = b + i;
    }
    swap(&a, &b);
    printf("a = %d, b = %d\n", a, b);
}
```

(i) $0 \leq 4$, $0/2 = 0$ false
Continue X

$a = -3 + 1 = -2$
 $b = -6 + 1 = -5$

(v) $4 \leq 4 \rightarrow$ true

$4/2 = 2$ false
Continue X

$a = -2 + 5 = 3$
 $b = -5 + 4 = -1$

(ii) $1 \leq 4$ $1/2 = 0$ false
Continue ✓

(iii) $2 \leq 4$ $2/2 = 1$ true
Continue X

$5 \leq 4$ false
 $\boxed{a = 6, b = 3}$

Gate 2006

Useless Printab()

```
void swap(int *x, int *y)
{
    static int *temp;
    temp = x;
    x = y; y = temp;
}
```

```
void printab() {
```

```
    static int i, a = -3, b = -6;
    i = 0;
```

```
    while(i <= 4) {
```

```
        if((i++) / 2 == 1) continue;
```

```
        a = a + i;
        b = b + i;
```

```
    }
    swap(&a, &b);
    printf("a = %d, b = %d\n", a, b);
}
```

(i) $i = 0$ $0 \leq 4 \rightarrow \text{true}$

$0 / 2 == 1$ false
continue x

$a = 7$
 $b = 4$

(ii) $1 \leq 4 \rightarrow \text{true}$

$1 / 2 == 1 \Rightarrow \text{true}$
continue

(iii) $2 \leq 4 \rightarrow \text{true}$

$2 / 4 == 1$ false
continue x

$a = 10$
 $b = 7$

(iv) $3 \leq 4 \rightarrow \text{true}$

$3 / 2 == 1$ true
continue;

(v) $4 \leq 4 \rightarrow \text{true}$

$4 / 2 == 1$ false
continue x

$a = 15$
 $b = 12$

$a = 6, b = 3$

```
void main() {
```

```
    i 0 1 2 3 4 5 printab();
```

```
    a 6 7 10 printab();
```

```
    b 3 4 7  
}
```